

Presentation Script

[Slide 1] - Intro

Hello! My name is Daniel and this is Duarte and welcome to our presentation.

[Slide 2] - The Problem

According to Agência Lusa, 72% of Portuguese people report struggling to distinguish between real and fake information on the internet. With this in mind, we trained several ML models to help mitigate this problem.

[Slide 3] - Dataset Overview

We used the FakeNews-PT dataset separated into training, validation, and test sets all of which we confirmed to be balanced.

[Slide 4] - Vocabulary Richness

Looking closer at the dataset we found that the vocabulary used in real news was significantly richer compared to fake news.

[Slide 5] - Average Word Length

We also examined word length, but didn't find any substantial differences.

[Slide 6] - Average Sentence Length

However, we found that, on average, the fake news articles contained longer sentences.

[Slide 7] - Text Preprocessing Strategies

We evaluated four preprocessing strategies, all of which include lowering and extra whitespace removal. We also applied punctuation removal, stop words removal and both of them combined.

[Slide 8] - Feature Extraction

For feature extraction, we utilized TF-IDF alongside IDF smoothing with the parameters set on the statement.

[Slide 9] - Model Training

When it comes to model training...

[Slide 10] - Model Hyperparameters

...we evaluated all five models with each preprocessing strategy. We conducted a grid search across the hyperparameter space that we defined, resulting in the configuration with the highest f1-score for each model.

[Slide 11] - Preprocessing Strategies Analysis

By averaging all five models across each preprocessing technique, we found that stop-word removal got the best overall results while punctuation removal actually degraded model performance.

[Slide 12] - Model Comparison

Our analysis from now on will be based on the stop-word removal strategy, since it proved to be the most accurate. With that established, when compared to the other models, MLP emerged as the one with the best results.

[Slide 13] - ROC Curve

The MLP's ability to distinguish between real and fake classes is illustrated by its ROC curve, having achieved an Area Under the Curve of 0.9779.

[Slide 14] - Confusion Matrix

The confusion matrix demonstrates residual class bias through a nearly symmetric error rate. This ensures identical precision and recall metrics that result in a highly stable F1-score. Furthermore, the probability distribution chart shows that the model is highly confident in its assignments, with predictions heavily clustered near 0 and 1.

[Slide 15] - Model Interpretation

When it comes to model interpretation...

[Slide 16] - Top Words Indicative of News Type

We extracted the most predictive words using Logistic Regression with L1 regularization and we found that words from the Old Orthographic Agreement like "acção" and "atividade" heavily predicted real news in the dataset.

[Slide 17] - Parts of Speech Impact

Looking at grammar weights, real news relied heavily on the subordinating and coordinating conjunctions that were not considered stop words, while fake news relied on numbers, adverbs, and verbs.

[Slide 18] - Entity Density

We also analyzed named entity density and found that fake news contained on average more references of locations and organizations, while real news showed a slightly higher density of personal names.

[Slide 19] - Regularization Comparison

When comparing L1 and L2 regularization, L1 turned irrelevant variables to zero, isolating the features that actually mattered, proving to be better to use for our high-dimensional and noisy data.

[Slide 20] - Feature Contribution

Furthermore, when comparing feature contributions across samples to compare Logistic Regression, which is linear, to MLP, which is non-linear, we can see that both of them identified a consistent set of words, but gave different weights to each of them.

[Slide 21] - Permutation Importance

We utilized permutation importance to confirm the MLP's global feature rankings, where tokens like "passos" and "região" turned out to be the primary drivers of model accuracy.

[Slide 22] - Clustering Analysis

We also utilized K-means clustering to group the articles into five different labels, effectively identifying key themes such as Domestic Politics, Economy, and Geopolitics.

[Slide 23] - Clusters vs Fake News

Plotting these K-Means clusters against the ground truth revealed that the "Curiosities & Clickbait" cluster was comprised of 77% fake news, which is the opposite to the 'Geopolitics' cluster at only 6%.

[Slide 24] - Preprocessing Experiments

To optimize classification results, we incorporated N-Grams to capture adjacent word relationships, and also applied lemmatization to reduce words to their base dictionary forms. These were applied to the Stop Words removal strategy, since it was the one with the best baseline results.

[Slide 25] - Performance with Advanced Preprocessing

However, these preprocessing experiments got no significant gains. While lemmatization provided a residual improvement, the introduction of N-grams resulted in a lower F1 score for the MLP model.

[Slide 26] - Exploring SVMs

We also reached out to NLP professor at IST Luísa Coheur, for her insights and she implementing SVM, which works by finding the most efficient boundary between classes.

[Slide 27] - Deep Learning with BERT

Another recommendation was fine-tuning a BERT model, which reads text in both directions, to capture subtle semantic patterns.

[Slide 28] - Final Model Heatmap

This heatmap illustrates that the SVM achieved a slightly inferior result, but still comparable to the MLP. While the BERT model outperformed all other traditional algorithms.

[Slide 29] - Conclusion

Thank you!